Solution

Here $a = 0, b = 0.5, n = 5, f(x) = x^2$

So, the width of each subinterval is

$$h = \Delta x = \frac{b-a}{n} = \frac{0.5-0}{5} = 0.1.$$

The partition of the interval is given by the points

$$x_0 = 0,$$

$$x_1 = x_0 + h = 0 + 0.1 = 0.1$$

$$x_2 = x_1 + h = 0.1 + 0.1 = 0.2$$

$$x_3 = x_2 + h = 0.2 + 0.1 = 0.3$$

$$x_4 = x_3 + h = 0.3 + 0.1 = 0.4$$

$$x_5 = x_4 + h = 0.4 + 0.1 = 0.5$$

(i) The left-end rule for Riemann sum with equal width Δx is

$$S = \left[f(x_0) + f(x_1) + \cdots + f(x_{n-1})\right]\Delta x.$$

$$\therefore S = \left[f(0) + f(0.1) + f(0.2) + f(0.3) + f(0.4)\right](0.1)$$

$$= \left[0.00 + 0.01 + 0.04 + 0.09 + 0.16\right](0.1) = 0.03$$

$\therefore \int_0^{0.5} x^2 dx$ is approximately 0.03.

(ii) The right-end rule for Riemann sum with equal width Δx is

$$S = \left[f(x_1) + f(x_2) + \cdots + f(x_n)\right]\Delta x.$$

$$\therefore S = \left[f(0.1) + f(0.2) + f(0.3) + f(0.4) + f(0.5)\right](0.1)$$

$$= \left[0.01 + 0.04 + 0.09 + 0.16 + 0.25\right](0.1) = 0.055.$$

$\therefore \int_0^{0.5} x^2 dx$ is approximately 0.055.

(iii)The mid-point rule for Riemann sum with equal width Δx is

$$S = \left[f\left(\frac{x_0 + x_1}{2}\right) + f\left(\frac{x_1 + x_2}{2}\right) + \cdots + f\left(\frac{x_{n-1} + x_n}{2}\right)\right]\Delta x$$

$$\therefore S = \left[f(0.05) + f(0.15) + f(0.25) + f(0.35) + f(0.45)\right](0.1)$$

$$= \left[0.0025 + 0.0225 + 0.0625 + 0.1225 + 0.2025\right](0.1)$$

$$= 0.04125.$$

$\therefore \int_0^{0.5} x^2 dx$ is approximately 0.04125.

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